

The Children of the World Rock: Apollo Lunar Sample 70017

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1 Introduction

The Apollo lunar rock 70017 has been one of the most widely studied of the returned lunar samples. This rock was called *The Children of the World Rock*; and subsequently, small pieces of the sample have been given to many countries for educational purposes.

2 Petrographic Observations

Paired rotational series of PPL and XPL images of sample 70017-r1 at 5° intervals was acquired from Virtual Microscope [1] and subsequently co-registered by using Python to ensure a consistent field of view. Observations were made on these co-registered PPL/XPL images. Phase identifications below rest on morphology plus optical behavior (relief, birefringence, twinning, extinction). Quantitative confirmation (EPMA, EBSD) is beyond the scope of these observations; assignments are stated as "consistent with" the named phase.

2.1 Skeletal opaque phase

Black, skeletal, opaque grains are abundant and widely distributed across the field. They are blocky to lath-shaped and stay black in both PPL and XPL at all stage rotations. The skeletal, embayed opaques are intergrown with, and partly enclosed by, the surrounding silicates. In reflected light, the opaques appear bright, as seen in Figure 2. Thus, an ilmenite-rich mode is expected.

2.2 Large, equant, subhedral grains

The most colorful grains in XPL display moderate-to-high, second-order interference colors, like browns and tans, blues, and patches of pink/magenta. Several grains show color variation from core to rim, the optical expression of compositional zoning. This matches the published description of pyroxene as large, zoned crystals trending from subcalcic augite through pigeonite to Fe-rich compositions, with titanium- and aluminum-rich cores [2–5]. In PPL, these same grains are pale brownish and traversed by faint cleavage. Individual pyroxenes are large enough to enclose plagioclase laths. Confident identification of these grains would require compositional analysis, so they are noted here descriptively rather than assigned.

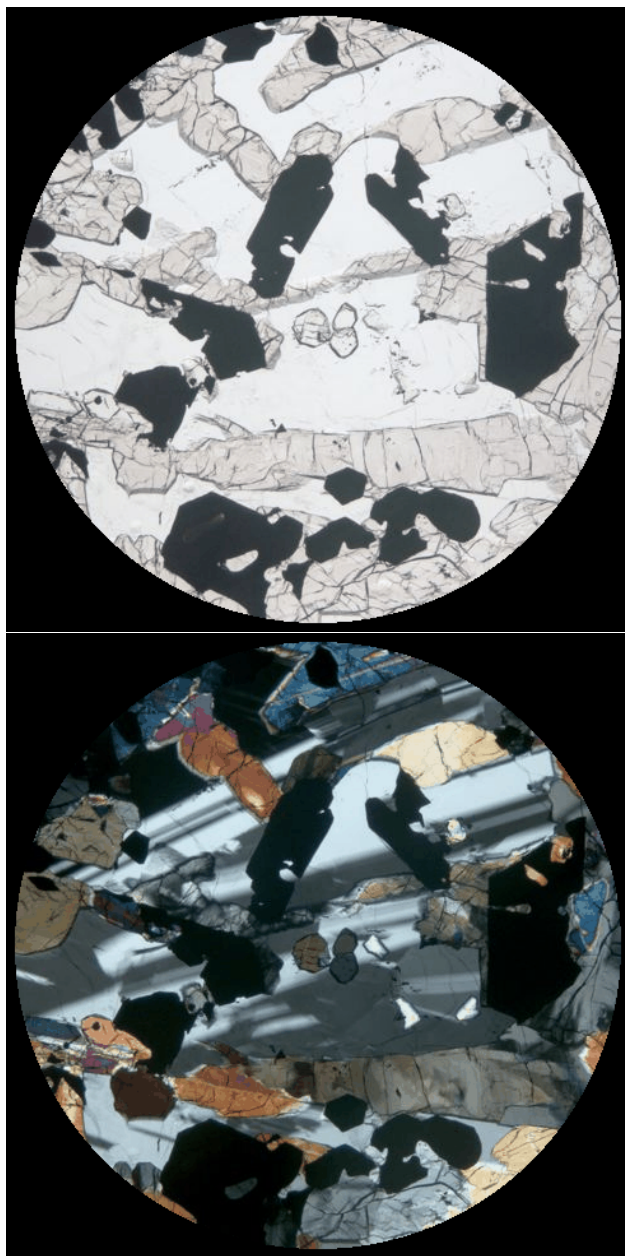


Figure 1: Sample 70017-r1 in PPL (top) and XPL (bottom). Field of view 2.0 mm. In PPL, opaque phases appear black and transparent silicates appear pale brown with relief and cleavage visible. In XPL, anisotropic silicates display interference colors and opaque phases remain black.

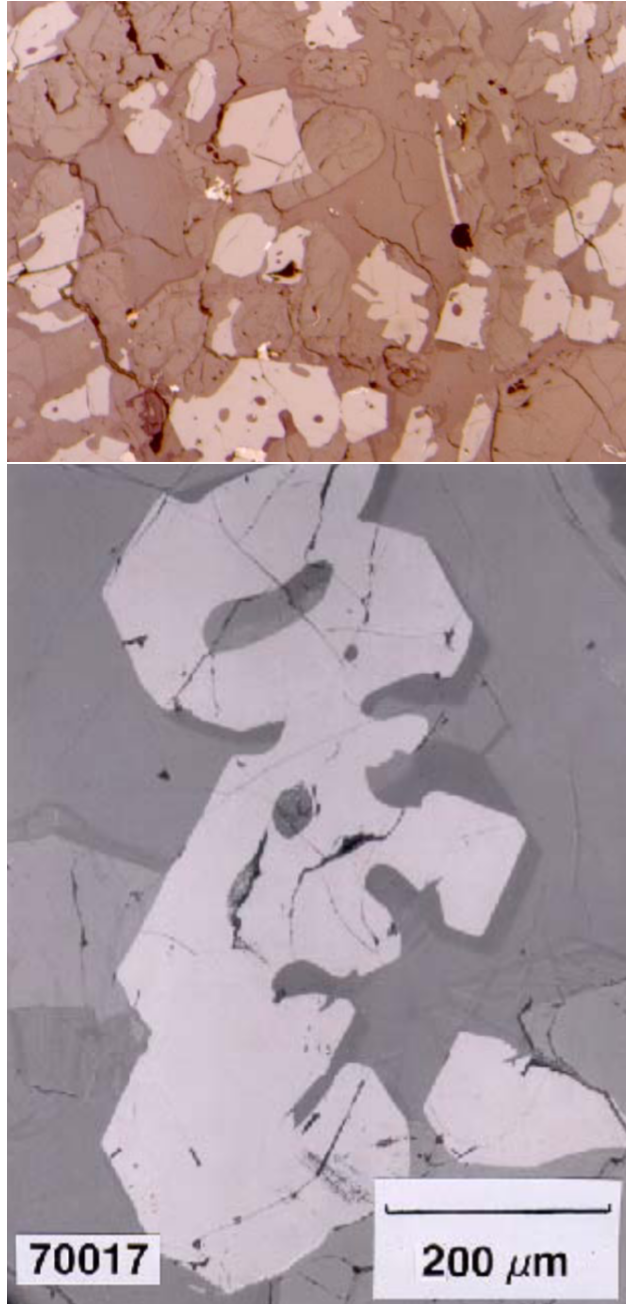


Figure 2: Sample 70017-r1 in reflected light. Credit: Lunar Sample Compendium.

2.3 Twinned laths

Lath-shaped grains showing grey-to-white, low-order interference colors and fine parallel (albite) twinning are consistent with plagioclase feldspar. They are subordinate in apparent area to pyroxene in this field and are commonly oriented at a range of angles, partially or wholly enclosed by the larger pyroxene grains.

A defining feature of this field is the relationship between pyroxene and plagioclase: large pyroxene grains, which go to extinction more or less as single optical units, enclose or wrap around multiple plagioclase laths. Where the laths are largely included within one pyroxene, the texture is poikilitic/ophitic; where they are only partly enclosed, it is subophitic. This is consistent with the reported occurrence of large poikilitic pyroxene and plagioclase in 70017 [2–5].

2.4 Minor olivine

Small euhedral grains (0.2mm) is present as euhedral grains poikilitically enclosed in twinned laths. They appear bright in PPL and display moderate to high interference colors, thus they are likely olivine.

2.5 Overall texture

The field is fully crystalline: there is no extinct glassy matrix, and essentially all of the area is occupied by interlocking mineral grains. Grain size is medium, with several crystals spanning a large fraction of the field. The fabric is a tightly interlocking, hypidiomorphic-granular to subophitic arrangement. This is the textural signature of comparatively slow, holocrystalline cooling.

3 Interpretation

The assemblage and textures in 70017-r1 record slow, near-complete crystallization of a high-Ti basaltic melt. Three lines of evidence point the same way: the medium, interlocking grain size; the absence of a glassy or skeletal-quench component; and the subophitic-to-poikilitic enclosure of plagioclase by large, well-developed pyroxene. Compositional zoning in the pyroxene tracks progressive crystallization of an evolving, increasingly Fe-rich residual liquid, while the abundant ilmenite reflects the high titanium content of the parent magma.

This cooling history is most consistent with crystallization in the interior of a relatively thick flow (or the dense core of a lava lobe), where heat was lost slowly enough to grow coarse, equilibrated crystals, in contrast to a chilled, glassy flow margin.

4 Conclusion

- 1) Thin section 70017-r1 is a medium, holocrystalline basalt composed of abundant opaque grains (consistent with ilmenite), large zoned pyroxene, and twinned plagioclase laths.
- 2) The pyroxene poikilitically to subophitically encloses plagioclase, and pyroxene interference colors record core-to-rim compositional zoning.
- 3) The texture indicates slow, near-complete crystallization, consistent with the flow-interior cooling history reported for this high-Ti ilmenite basalt.

References

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- [2] C. Meyer, *Lunar Sample Compendium*. Houston, TX, USA: NASA Johnson Space Center, 2011. [Online]. Available: <https://curator.jsc.nasa.gov/lunar/lsc/>
- [3] J. Longhi, D. Walker, E. Stolper, T. L. Grove, and J. F. Hays, "Petrology and crystallization history of basalts 70017 and related Apollo 17 mare basalts," in *Proc. 5th Lunar Sci. Conf.*, Houston, TX, USA, 1974.
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